

Total No. of Printed Pages: 3

B.E - (Mechanical) (Semester-VII)(Revised Course 2019-2020)**OME725 OPEN ELECTIVE Additive Manufacturing****[Time: 3:00 Hours]****[Max. Marks:100]**

- Instructions:**
- 1) Answer **any two** questions each from **Part A** and **Part B** and **one** question from **part C**
 - 2) Assume missing data, if any
 - 3) illustrate your answers with neat sketches wherever required.

PART AAnswer any **TWO** questions from the following:

- Q1 (A) A company that produces customized medical devices needs to create detailed plastic parts with high dimensional accuracy. How does the Solid Ground Curing (SGC) process ensure that the required precision is achieved? **07**
- (B) What are the different classifications of additive manufacturing technologies based on form of raw material, and how do these classifications influence the choice of process for specific applications? **07**
- (C) What are the key differences between additive manufacturing and subtractive manufacturing in terms of cost, material usage, and design flexibility? What are the strengths and weaknesses of each approach? **06**
- Q2 (A) Which liquid-based additive manufacturing system is most suitable for a medical device startup that needs to manufacture highly detailed surgical guides with complex geometries for orthopedic surgeries, and why? **07**
- (B) A consumer electronics company aims to produce lightweight, complex housing prototypes for their latest product using 1.75 mm diameter PLA filament. What suitable additive manufacturing process should they utilize? Please describe the working principle and outline the advantages and disadvantages of this process. Include a neat sketch if possible. **07**
- (C) Which powder-based additive manufacturing process is most suitable for an aerospace company looking to manufacture high-performance, lightweight components that require high accuracy, strength, and density? **06**

- Q3 (A) What is the significance of additive manufacturing in modern product development, and what key advantages does it offer over traditional manufacturing methods? **07**
- (B) An industrial equipment manufacturer needs to repair a damaged high-strength metal component in a cost-effective manner. Which powder-based additive manufacturing method would be most suitable for this application? **07**
- (C) What are the various applications of additive manufacturing across different industries? Provide examples of how this technology is transforming product design and manufacturing. **06**

PART B

Answer any **TWO** questions from the following:

- Q4 (A) An electronics company needs to rapidly prototype detailed circuit board casings with smooth surface finishes. How would a thermal jet printer-based 3D modeler be suitable for this application, and what are its limitations? **07**
- (B) company needs to produce a small batch of customized injection molds quickly. Which indirect rapid tooling method would be most cost-effective and why? **07**
- (C) A manufacturer is looking to digitize their existing physical parts for reverse engineering. What digitization technique would be most suitable for capturing detailed geometric data? **06**
- Q5 (A) A consumer products company needs to rapidly prototype detailed packaging designs with smooth surface finishes. How would a thermal jet printer-based 3D modeller be suitable for this application, and what are its limitations? **07**
- (B) A toy manufacturer needs to produce a small batch of customized moulds for new product lines quickly. Which indirect rapid tooling method would be most cost-effective for this purpose, and why? **07**
- (C) An automotive company is looking to digitize their existing physical components for reverse engineering. What digitization technique would be most suitable for capturing detailed geometric data? **06**
- Q6 (A) A manufacturer notices significant dimensional inaccuracies in parts built using additive manufacturing. How can optimizing build orientation help reduce errors in the final parts? **07**

- (B) In the field of computer-aided tissue engineering (CATE), how does bio-additive manufacturing contribute to the development of functional tissue constructs? **07**
- (C) A designer is preparing a complex 3D model for 3D printing and needs to save it in the correct format. What are the key characteristics of STL files that make them suitable for additive manufacturing? **06**

PART C

(Answer any ONE question from the following)

- Q7 (A) A company is considering options for producing a limited run of custom parts. What is one primary advantage of using soft tooling over hard tooling in this scenario? **07**
- (B) A researcher is studying case studies in bio-additive manufacturing to understand the challenges faced in creating customized prostheses. What common challenges might they encounter in the design and production phases? **07**
- (C) Write a note on Rapid manufacturing process optimization **06**
- Q8 (A) Compare the suitability of Stereolithography (SLA) and Fused Deposition Modeling (FDM) for producing functional prototypes. What factors should be considered when selecting between the two methods? **07**
- (B) A jewelry manufacturer is evaluating the use of a thermal jet printer-based system for creating small, highly detailed prototypes. What features of this system make it suitable for such applications? **07**
- (C) A packaging design firm is seeking an additive manufacturing process that can create prototypes using biodegradable materials. What process would you recommend, and what are its working principles, advantages, and disadvantages? Include a neat sketch if possible. **06**